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Root Systems and the Representation of Lie Algebras

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Abstract. This paper develops the representation theory of complex semisimple Lie algebras from first principles, culminating in the Cartan–Killing classification of simple Lie algebras via root systems and Dynkin diagrams. Beginning with the foundational algebraic structure of a Lie algebra and its adjoint representation, the paper develops the theory of weights and highest weight vectors, and states the Theorem of the Highest Weight governing finite-dimensional irreducible representations. The Cartan subalgebra and the associated root space decomposition are then introduced, with the additivity of roots derived directly from the Jacobi identity. The geometric axioms of an abstract root system are stated and used to derive the discrete set of angles permissible between roots, motivating the construction of simple root systems, Cartan matrices, and the Weyl group. The paper concludes by presenting the Dynkin diagram formalism and the resulting classification of every finite-dimensional complex simple Lie algebra into one of four classical families or five exceptional algebras.

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1. Preliminaries

This section establishes the foundational definitions of Lie algebras, stripping away the continuous topology of Lie groups to focus entirely on their linearized, algebraic structure.

1.1 Lie Algebra

A Lie algebra is a vector space $\mathfrak{g}$ over a field F (in this paper, we assume $F = \mathbb{C}$), equipped with a binary operation $[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$ called the Lie bracket. To qualify as a Lie bracket, the operation must satisfy three axioms for all $x, y, z \in \mathfrak{g}$ and scalars $a, b \in F$:

1. **Bilinearity:** $[ax + by, z] = a[x, z] + b[y, z]$ and $[z, ax + by] = a[z, x] + b[z, y]$.

2. **Alternativity:** $[x, x] = 0$. (When the characteristic of F is not 2, this implies strict anti-commutativity: $[x, y] = -[y, x]$).

3. **The Jacobi Identity:** $[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0$.

1.2 Lie Algebra Homomorphism

A linear transformation $\phi : \mathfrak{g} \to \mathfrak{h}$ between two Lie algebras is a homomorphism if it strictly preserves the Lie bracket structure. That is, for all $x, y \in \mathfrak{g}$:

$$\phi([x, y]_{\mathfrak{g}}) = [\phi(x), \phi(y)]_{\mathfrak{h}}$$

This structural preservation can be visualized using a commutative diagram, which asserts that applying the bracket in $\mathfrak{g}$ then mapping to $\mathfrak{h}$ yields the identical result to mapping the elements first, then applying the bracket in $\mathfrak{h}$.

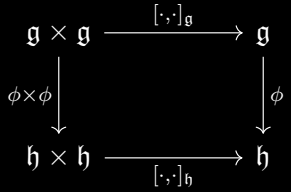


Figure 1: Commutative diagram of a Lie algebra homomorphism.

1.3 Lie Subalgebras and Ideals

A subspace $\mathfrak{h} \subseteq \mathfrak{g}$ is a Lie subalgebra if it is closed under the Lie bracket: $[\mathfrak{h}, \mathfrak{h}] \subseteq \mathfrak{h}$. A subspace $\mathfrak{i} \subseteq \mathfrak{g}$ is an ideal if it satisfies a much stronger condition, absorbing multiplication from the entire algebra such that $[\mathfrak{g}, \mathfrak{i}] \subseteq \mathfrak{i}$. Ideals in Lie algebras serve the exact same structural decomposition role as normal subgroups do in finite group theory.

1.4 The Adjoint Representation

This is the most critical definition for the remainder of the paper. For every element $x \in \mathfrak{g}$, we define a linear map $\mathrm{ad}_x : \mathfrak{g} \to \mathfrak{g}$ given by:

$$\mathrm{ad}_x(y) = [x, y]$$

The mapping $\mathrm{ad} : \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g})$ (where $\mathfrak{gl}(\mathfrak{g})$ is the general linear Lie algebra of all endomorphisms of $\mathfrak{g}$) is itself a Lie algebra homomorphism. This map allows the abstract algebra to act upon itself via concrete matrices.

2. Representations, Modules, and Highest Weights

2.1 Representation of a Lie Algebra

Let V be a finite-dimensional vector space over $\mathbb{C}$. A representation of a Lie algebra $\mathfrak{g}$ is a Lie algebra homomorphism $\rho : \mathfrak{g} \to \mathfrak{gl}(V)$, where $\mathfrak{gl}(V)$ is the general linear algebra of endomorphisms on V.
Because ρ is a homomorphism, it must strictly preserve the bracket structure:

$$\rho([x, y]) = \rho(x)\rho(y) - \rho(y)\rho(x) \quad \text{for all } x, y \in \mathfrak{g}$$

2.2 Lie Algebra Module

The language of representations is completely equivalent to the language of modules. A vector space V is a $\mathfrak{g}$-module if there exists a bilinear operation $\cdot : \mathfrak{g} \times V \to V$ satisfying:

$$[x, y] \cdot v = x \cdot (y \cdot v) - y \cdot (x \cdot v) \quad \text{for all } x, y \in \mathfrak{g}, \text{ and } v \in V$$

This equivalence can be represented categorically. If we let $\mathrm{End}(V)$ denote the endomorphism ring of V, the module action defines a map from $\mathfrak{g}$ to $\mathrm{End}(V)$ that commutes with the bracket operation.

$$\begin{array}{ccc} \mathfrak{g} \times \mathfrak{g} & \xrightarrow{\quad [\cdot,\cdot] \quad} & \mathfrak{g} \\ \Big\downarrow{\scriptstyle \rho \times \rho} & & \Big\downarrow{\scriptstyle \rho} \\ \mathfrak{gl}(V) \times \mathfrak{gl}(V) & \xrightarrow[\text{commutator}]{} & \mathfrak{gl}(V) \end{array}$$

Figure 2: The representation map ρ respects the algebraic bracket by translating it to a matrix commutator in $\mathfrak{gl}(V)$.

2.3 Weights and Weight Spaces

To dissect a representation (V, ρ), we rely on simultaneous diagonalization. We select a Cartan subalgebra $\mathfrak{h} \subset \mathfrak{g}$ (a maximal abelian subalgebra). Because the elements of $\mathfrak{h}$ commute with one another, their matrix representations $\rho(h)$ can be simultaneously diagonalized on V.

Instead of single scalar eigenvalues, the generalized eigenvalues of $\mathfrak{h}$ are linear functionals mapping $\mathfrak{h}$ to $\mathbb{C}$. These functionals are called weights.

Let $\mathfrak{h}^*$ be the dual space of $\mathfrak{h}$. A linear functional $\lambda \in \mathfrak{h}^*$ is called a weight of V if there exists a non-zero vector $v \in V$ such that:

$$h \cdot v = \lambda(h)v \quad \text{for all } h \in \mathfrak{h}$$

The subspace of all such vectors is the weight space V_λ. The entire module V decomposes into the direct sum of its weight spaces: $V = \bigoplus_\lambda V_\lambda$.

2.4 Highest Weight Vector

A non-zero vector $v \in V_\lambda$ is a highest weight vector if it is annihilated by the action of all positive root spaces in $\mathfrak{g}$. That is, $e_\alpha \cdot v = 0$ for all positive roots $\alpha \in \Phi^+$.

2.5 Highest Weight Theorem

For every complex finite-dimensional semisimple Lie algebra:

1. Every finite-dimensional irreducible representation contains a unique (up to scalar multiplication) highest weight vector.
2. Two irreducible representations are isomorphic if and only if they possess the same highest weight.
3. For every "dominant integral" weight $\lambda \in \mathfrak{h}^*$, there exists exactly one unique finite-dimensional irreducible representation that has λ as its highest weight.

3. The Cartan Subalgebra and Root Space Decomposition

3.1 The Cartan Subalgebra

A Cartan subalgebra $\mathfrak{h}$ of a semisimple Lie algebra $\mathfrak{g}$ is a maximal abelian subalgebra consisting entirely of semisimple elements (elements whose adjoint representations are diagonalizable).

Because $\mathfrak{h}$ is abelian ($[h_1, h_2] = 0$ for all $h_1, h_2 \in \mathfrak{h}$), the set of linear operators $\{\text{ad}_h \mid h \in \mathfrak{h}\}$ is a commuting family of diagonalizable matrices. A foundational theorem in linear algebra dictates that a commuting family of diagonalizable matrices can be simultaneously diagonalized.

3.2 Roots and Root Spaces

Because we can simultaneously diagonalize the action of $\mathfrak{h}$ on $\mathfrak{g}$, the entire vector space $\mathfrak{g}$ decomposes into eigenspaces.
For any linear functional $\alpha \in \mathfrak{h}^*$ (where $\mathfrak{h}^*$ is the dual space of $\mathfrak{h}$), we define the root space $\mathfrak{g}_\alpha$ as:

$$\mathfrak{g}_\alpha = \{x \in \mathfrak{g} \mid [h, x] = \alpha(h)x \text{ for all } h \in \mathfrak{h}\}$$

If $\alpha \neq 0$ and $\mathfrak{g}_\alpha$ contains non-zero vectors, then α is called a root of $\mathfrak{g}$ relative to $\mathfrak{h}$. The set of all roots is denoted by Φ.
The non-zero elements of $\mathfrak{g}_\alpha$ are called root vectors.

3.3 The Cartan Decomposition

The Lie algebra $\mathfrak{g}$ decomposes into a direct sum of its Cartan subalgebra and its one-dimensional root spaces. This is the Root Space Decomposition:

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_\alpha$$

The zero eigenspace is the Cartan subalgebra itself: $\mathfrak{g}_0 = \mathfrak{h}$. For semisimple Lie algebras, it is a critical structural fact that every root space $\mathfrak{g}_\alpha$ is exactly one-dimensional.

3.4 Additivity of Roots

If α and β are roots, and $x \in \mathfrak{g}_\alpha$, $y \in \mathfrak{g}_\beta$, then $[x, y] \in \mathfrak{g}_{\alpha+\beta}$.
Proof: Let $h \in \mathfrak{h}$. Using the Jacobi identity and the anti-commutativity of the bracket:

$$[h, [x, y]] = [[h, x], y] + [x, [h, y]]$$
$$[h, [x, y]] = [\alpha(h)x, y] + [x, \beta(h)y]$$
$$[h, [x, y]] = (\alpha(h) + \beta(h))[x, y]$$

This confirms that $[x, y]$ is an eigenvector for ad_h with eigenvalue $\alpha(h) + \beta(h)$.
Thus, the Lie bracket maps $\mathfrak{g}_\alpha \times \mathfrak{g}_\beta \to \mathfrak{g}_{\alpha+\beta}$. If $\alpha + \beta$ is not a root (and is not zero), then $[x, y] = 0$.

4. The Geometry of Root Systems

4.1 Root System

A subset Φ of a Euclidean space E is a root system if it satisfies four geometric axioms:

1. **Finiteness and Span:** Φ is finite, does not contain the zero vector, and spans E.

2. **Integrality of Projections:** If α and β are in Φ, the quantity twice the inner product of β and α, divided by the squared length of α — equivalently, the length of the orthogonal projection of β onto α, measured in units of half the length of α — must be an integer. This is the Cartan integer, denoted as $\langle \beta, \alpha \rangle$:

$$\langle \beta, \alpha \rangle = 2\frac{(\beta, \alpha)}{(\alpha, \alpha)} \in \mathbb{Z}$$

3. **Reflection Symmetry:** If $\alpha \in \Phi$, the reflection of the entire system Φ across the hyperplane orthogonal to α leaves the system unchanged. The reflection operator s_α is defined as:

$$s_\alpha(\beta) = \beta - \langle \beta, \alpha \rangle \alpha$$

   For any $\beta \in \Phi$, the reflected vector $s_\alpha(\beta)$ is also in Φ.

4. **Proportionality:** If $\alpha \in \Phi$ and $c\alpha \in \Phi$ for some scalar c, then $c = \pm 1$. The only scalar multiples of a root that are also roots are the root itself and its negative.

These axioms are incredibly restrictive. The integrality condition (Axiom 2) severely limits the possible angles θ between any two roots α and β. Because $4\cos^2\theta = \langle \alpha, \beta \rangle \langle \beta, \alpha \rangle$, and this product must be an integer between 0 and 4, the only permissible angles between roots are 90°, 120° (60°), 135° (45°), and 150° (30°).

4.2 Simple Roots (Δ)

A subset $\Delta = \{\alpha_1, \alpha_2, \ldots, \alpha_n\} \subset \Phi$ is a base of simple roots if:

1. Δ forms a vector space basis for E.

2. Every root $\beta \in \Phi$ can be written as a linear combination of simple roots, $\beta = \sum k_i \alpha_i$, where the coefficients k_i are integers that are either all non-negative or all non-positive.

4.3 The Cartan Matrix

Let $\Delta = \{\alpha_1, \ldots, \alpha_n\}$ be a set of simple roots. The Cartan Matrix A is an $n \times n$ matrix whose entries A_{ij} are the Cartan integers:

$$A_{ij} = \langle \alpha_i, \alpha_j \rangle = 2\frac{(\alpha_i, \alpha_j)}{(\alpha_j, \alpha_j)}$$

The diagonal entries are always 2 (since $\langle \alpha_i, \alpha_i \rangle = 2$). The off-diagonal entries are negative integers or zero, dictating exactly the angles and relative lengths between the simple roots. From this single matrix, one can generate the entire root system Φ, the Weyl group, and uniquely identify the original Lie algebra up to isomorphism.

4.4 The Weyl Group (W)

The Weyl group W is the subgroup of the orthogonal group of E generated by the root reflections s_α for all $\alpha \in \Phi$.
Because the root system is finite and spans E, W must be a finite group. It acts faithfully on the set of roots Φ, permuting them while preserving the geometric structure of the lattice.

Geometrically, the hyperplanes orthogonal to the roots slice the Euclidean space E into distinct, open regions called Weyl chambers. The choice of a set of simple roots Δ is geometrically equivalent to selecting a single, fundamental Weyl chamber. The Weyl group W acts simply transitively on these chambers—meaning that applying the Weyl group elements to the fundamental chamber perfectly tiles the entire space without overlap.

5. Dynkin Diagrams and the Classification Theorem

The Cartan matrix contains all the information needed to reconstruct a semisimple Lie algebra, but working with matrices of integers is cumbersome. In 1947, Eugene Dynkin introduced a brilliant graph-theoretic shorthand that visually encodes the Cartan matrix. These are known as Dynkin diagrams.

Let $\Delta = \{\alpha_1, \alpha_2, \ldots, \alpha_n\}$ be a base of simple roots for a root system Φ. The Dynkin diagram is a graph constructed by the following strict rules:

1. **Nodes:** Draw one node (vertex) for each simple root $\alpha_i \in \Delta$. The number of nodes is exactly the rank of the Lie algebra.

2. **Edges:** The number of edges connecting node i to node j is determined by the product of their Cartan integers: $n_{ij} = \langle \alpha_i, \alpha_j \rangle \langle \alpha_j, \alpha_i \rangle$.
   - $n_{ij} = 0 \implies$ No edge (roots are orthogonal, 90°).
   - $n_{ij} = 1 \implies$ Single edge (roots are at 120°).
   - $n_{ij} = 2 \implies$ Double edge (roots are at 135°).
   - $n_{ij} = 3 \implies$ Triple edge (roots are at 150°).
3. **Arrows:** If two roots are connected by multiple edges (meaning they have different lengths), draw an arrow pointing from the longer root to the shorter root.

5.1 The Classification Theorem

A Lie algebra is "simple" if its root system cannot be partitioned into two mutually orthogonal subsets. In the language of Dynkin diagrams, a Lie algebra is simple if and only if its Dynkin diagram is a connected graph. By exhaustively analyzing which connected graphs satisfy the geometric axioms of a root system (specifically the requirement that the corresponding Cartan matrix is positive-definite), mathematicians achieved one of the most monumental results in 20th-century algebra: The Cartan-Killing Classification. Every finite-dimensional complex simple Lie algebra corresponds to exactly one of four infinite families, or one of five isolated "exceptional" cases.

5.2 The Four Classical Families

These families correspond to the standard symmetries of vector spaces that you encounter in linear algebra.

1. A_n ($n \geq 1$): A simple chain of n nodes.
   - **Algebra:** The special linear algebra $\mathfrak{sl}_{n+1}(\mathbb{C})$ (matrices with trace zero).
2. B_n ($n \geq 2$): A chain of nodes ending in a double edge with an arrow pointing right.
   - **Algebra:** The odd-dimensional special orthogonal algebra $\mathfrak{so}_{2n+1}(\mathbb{C})$ (skew-symmetric matrices).
3. C_n ($n \geq 3$): A chain of nodes ending in a double edge with an arrow pointing left.

- **Algebra:** The symplectic algebra $\mathfrak{sp}_{2n}(\mathbb{C})$ (matrices preserving a non-degenerate skew-symmetric form).

4. D_n ($n \geq 4$): A chain of nodes that forks at the very end into two orthogonal roots.

    - **Algebra:** The even-dimensional special orthogonal algebra $\mathfrak{so}_{2n}(\mathbb{C})$.

5.3 The Five Exceptional Algebras

The geometric constraints allow for exactly five highly complex configurations that do not fit into the standard matrix families. These are the exceptions that prove the rule.

- G_2: Two nodes connected by a triple edge. This is the algebra of derivations of the octonions (dimension 14).

- F_4: Four nodes, with a double edge in the middle. (Dimension 52).

- E_6, E_7, E_8: These are formed by taking the A_n chain and attaching a single node to the third vertex from the end. The largest, E_8, is a 248-dimensional beast whose associated root lattice provides the densest possible sphere packing in eight dimensions (a result proven by Viazovska in 2016), and which appears as the gauge symmetry algebra of one of the two consistent heterotic string theories, $E_8 \times E_8$.

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